Assignment-3

What to Hand In

You should upload a pdf or word file for **Problem-1**, **Problem-2** and **Problem-3**.

For the other problems, use the provided assignment link.

Problem-4 (6 points):

Consider the given class Attendance (p4.cpp). The class contains:

- A private string data member attendance which represents an attendance record for a student where each character signifies whether the student was absent, late, or present on that day. The record only contains the following three characters:
  - 'A': Absent.
  - 'L': Late.
  - 'P': Present.
- A private method isAbsent that returns true if student missed 2 or more days and false otherwise.
- A private method isLate that returns true if student was late for 3 or more consecutive days, and false otherwise.
- A constructor that constructs an Attendance object with a specified attendance record.
- A method checkAttendance that returns true if student was absent for strictly fewer than 2 days total **and** was never late for 3 or more consecutive days.

Finish the implementation of the class methods. **Do not modify the given test program.**

Hint: use string function *find* in isLate method.

Problem-5 (6 points):

Design a class named Stock (do not separate class definition with class implementation) that contains the following:

- A private string data field named symbol for the stock's symbol.
- A private string data field named name for the stock's name.
- A private double data field named previousClosingPrice that stores the stock price for the previous day.
- A private double data field named currentPrice that stores the stock price for the current time.

- A constructor that creates a stock with specified symbol and name.
- The constant accessor functions for all data fields.
- The mutator functions for previousClosingPrice and currentPrice.
- A constant function named getChangePercent() that returns the percentage changed from previousClosingPrice to currentPrice.

Do not modify the given test program.

Here is a sample run:

```
Previous Closing Price: 27.5
Current Price: 27.6
Percentage Change: 0.00363636
```

Problem-6 (7 points):

Description:

You have recently been hired by **FloridaPolytesla** company, a new rocket manufacturer, and have been tasked with designing a simulator for their spaceships. This simulator creates a virtual "space" and keeps track of the basic movements and direction of a given ship. Your job is to keep track of where the ship is and its orientation relative to its starting point.

Your ship simulator uses a string that represent a planned flight path for a given rocket ship.

In a ship's flight path, there are only 3 valid options for movement: **R** for turning right, **L** for turning left and **A** for advancing.

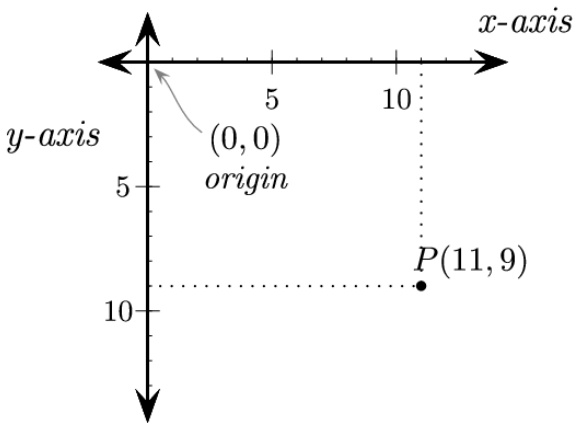
If, for example, you receive "RRALAA" as your flight path, you should interpret it as the following: Turn right, turn right, advance, turn left, advance, advance. Your program should calculate the final destination of a spaceship as well as it's orientation relative to the starting point.

Orientation is represented as **left**, **right**, **up** or **down**.

All spaceships will start at 0,0 and will face up.

Space is infinite, so the x, y coordinates you return could be placed on a seemingly infinite grid and can be negative or positive values.

We are using Computer Graphics Coordinate System.



Instructions:

Create a class named Spaceship that contains the following:

- A private int data **x**
- A private int data **y**
- A private string data named **position** which has the following format:
  - "{x: X, y: Y, direction: 'DIRECTION'}"
  - **X**,**Y** represent the ending coordinates of your ship and **direction** represents its final direction.
- No argument constructor which sets x and y coordinates to 0 and initializes the position with "{x: 0, y:0, direction: 'up'}"
- Constructor with the following prototype
  - **Spaceship(const std::string& path);**
  - Constructor takes in a flight path of a rocket ship as a string, calculates the final position of the spaceship and initializes position data member of the spaceship object. *Tip: google to_string().*
- The constant accessor function named **getPosition()** which returns the position of the spaceship.

You are given **testSpaceship.cpp** file. Implement the class in **Spaceship.h** and **Spaceship.cpp** files.

Here is the expected output:

```
{x: 0, y:0, direction: 'up'}
{x: 2, y: -1, direction: 'down'}
{x: 7, y: -5, direction: 'right'}
{x: 0, y: 0, direction: 'up'}
{x: -3, y: 0, direction: 'left'}
```